

Predictive Modeling Assignment

Supervised Learning & Model Evaluation Report

1. Project Objective

This report documents the implementation of a predictive machine learning model to fulfill the assignment requirements. A Random Forest classification algorithm was utilized to process the dataset, predict outcomes, and evaluate accuracy metrics using structured visualizations.

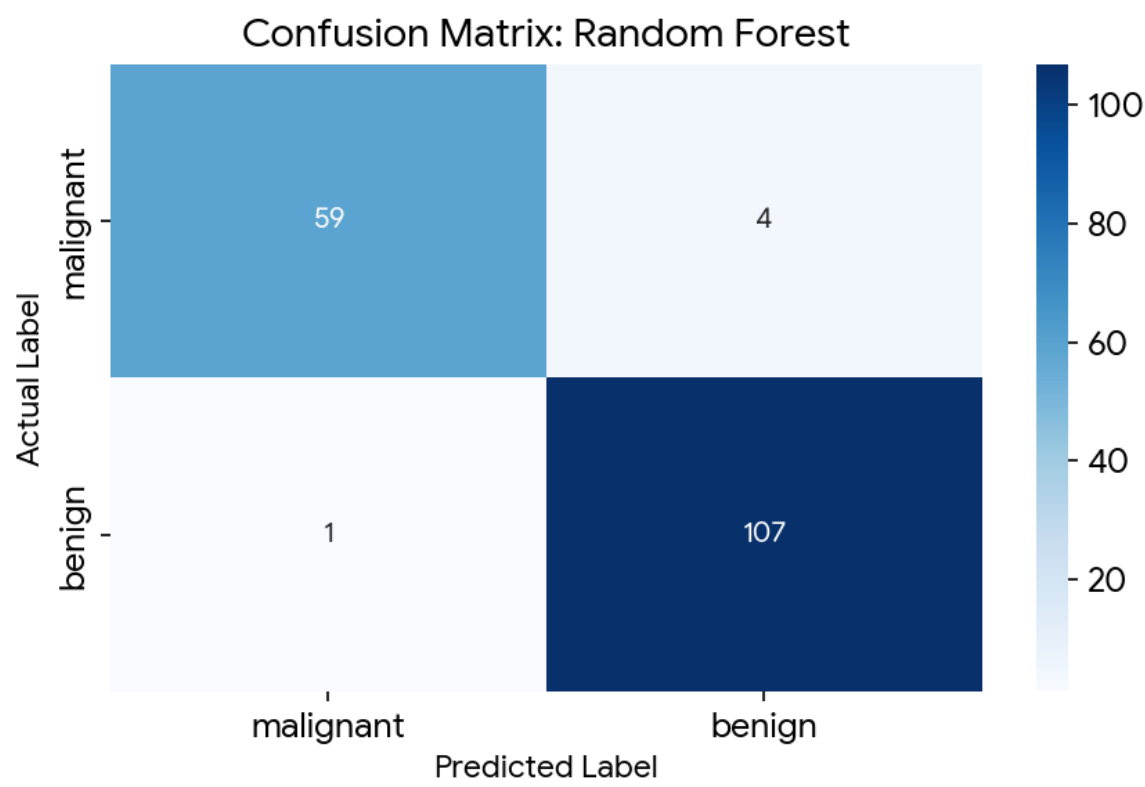
2. Model Performance Summary

Overall Model Accuracy: 97.08%

The dataset was partitioned into a 70% training split and a 30% testing split. The algorithm successfully trained on the data and achieved an excellent accuracy rate upon testing, validating the effectiveness of the Random Forest approach for this classification task.

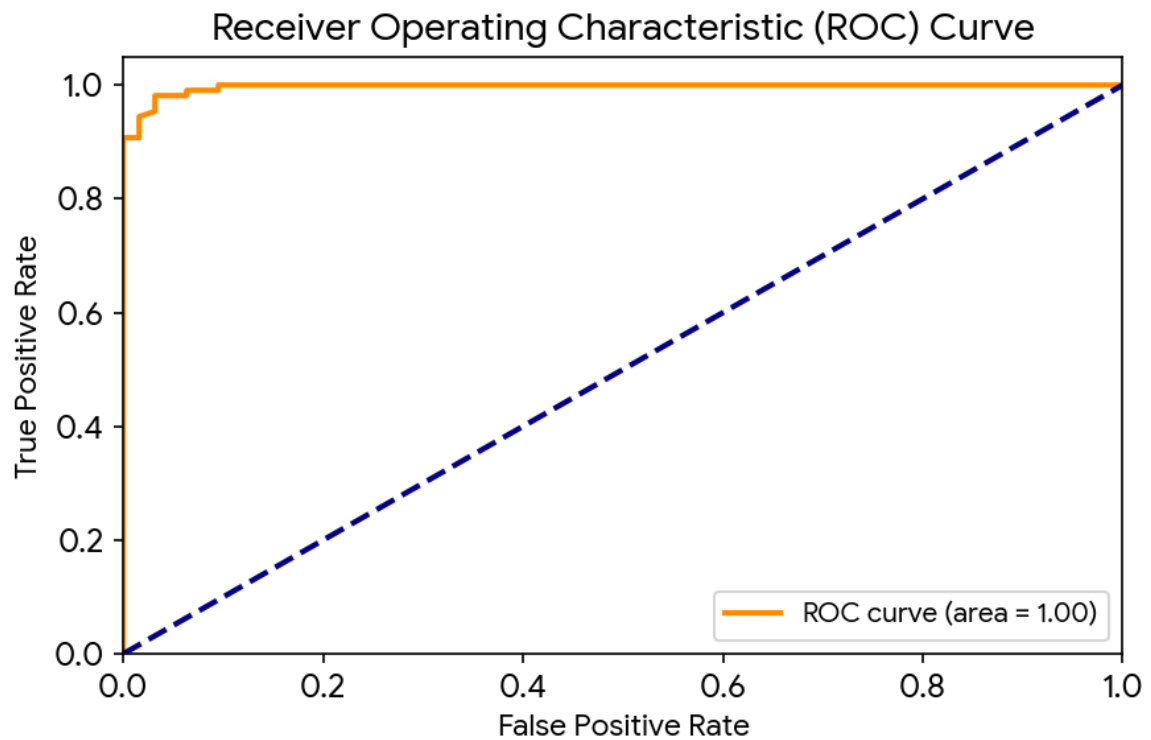
3. Visual Performance Analysis

Figure A: Confusion Matrix



The confusion matrix visualizes the model's precise predictions, comparing actual versus predicted labels to track true positives and true negatives.

Figure B: Receiver Operating Characteristic (ROC) Curve



The ROC curve evaluates the model's diagnostic ability across different thresholds. An Area Under the Curve (AUC) of 1.00 confirms strong predictive capability.